

Name of the Student _____

Scholar Number _____

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
MID TERM EXAMINATION-MARCH 2024

Course: B.Tech

Semester: IV

Subject Name: POWER SYSTEM

Subject Code: EE-221

Time Duration: 1 Hour 30 Min

Maximum Marks: 20

Subject Coordinator Name: Dr. Yogendra Kumar and Dr. More Raju

Note: Assume necessary data if required.

S.No	Question	Marks	CO
1	A) A 50 Hz, 3-phase line consist of three equal conductors with radius of each conductor 12.5 cm, placed in horizontal plane, with a spacing of 6 m between the middle and each outer conductor. Determine the inductive and capacitive reactance per phase per km of the transposed line.	3	CO-2
	B) A 50 km long line supplies a load of 5 MVA at 0.8 power factor lagging at 33 kV. The efficiency of transmission is 90 percent. Calculate the volume of conductor (aluminum) required for the line when, (i) Single-phase, two wire system is used. (ii) Three-phase, three wire system is used.	2	CO-2
2	A) Explain how the most economical cable size is determined.	2	CO-3
	B) A 60kV (rms) single core metal sheathed cable is to be graded by means of a metallic inter-sheath. The safe electric stress of the insulating material is 4kV/mm (rms). (i) Calculate the diameter of the inter sheath and the voltage at which it must be maintained in order to obtain minimum overall diameter. Also calculate corresponding conductor diameter. (ii) Compare the conductor obtained with that of an ungraded cable working under the same conditions.	3	CO-3
3	A) A three phase 50 Hz transmission line has resistance, inductance and capacitance per phase of 10 Ω , 0.1 H and 0.9 μF respectively and delivers a load of 35 MW at 132 kV and 0.8 p.f. lag. Determine the efficiency and regulation of the line using (i) nominal-T (ii) nominal- π methods.	4	CO-3
	B) A 150 km, 3-phase, 110 kV, 50 Hz transmission line transmits a load of 40 MW at 0.9 pf lag at receiving end. Resistance/km/phase = 0.15 Ω , reactance/km/phase = 0.6 Ω , susceptance/km/phase = 10^{-5} S. Determine A, B, C, and D constants and regulation of line.	2	CO-3
4	A) A string of 5 insulator units has a self-capacitance equal to 10 times the pin-to-earth capacitance. Find the voltage distribution across various units as a percentage of total voltage across the	4	CO-3